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Predictive Maintenance in Smart Manufacturing: Evaluating AI Models for Equipment Failure Prediction

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Prepared by:

Amaan Mohammed Khalander

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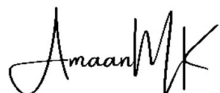
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AMAAN MOHAMMED KHALANDER

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Executive Summary

This essay investigates AI-driven predictive maintenance (PdM) for smart manufacturing, focusing on sensor-based fault detection and Remaining Useful Life (RUL) estimation. Drawing on twelve peer-reviewed sources (2021–2025), it evaluates methodologies, datasets, and key gaps.

Two models are compared: Long Short-Term Memory (LSTM) networks and Temporal Convolutional Networks (TCNs). The essay covers data requirements, preprocessing, experimental design, and a critical analysis of interpretability, scalability, and deployment. It concludes that a hybrid TCN–tree-based strategy is the most industrially defensible approach.

1. Introduction

Unplanned downtime imposes major costs on manufacturers. Reactive and preventive strategies are increasingly inadequate in Industry 4.0, where sensor networks generate continuous vibration, temperature, pressure, and acoustic data. Predictive maintenance (PdM) uses AI models to forecast failures before they occur, enabling proactive scheduling and cost reduction (Bampoula et al., 2024).

In the case study, a smart manufacturing company operates pumps, motors, and conveyor systems, each instrumented at high frequency. Management requires a solution to estimate the RUL of each component and flag anomalies ahead of breakdowns. This essay addresses all five brief tasks: literature review, model selection, preprocessing, experimental design, and critical analysis of deployment.

2. Literature Review (2021–2025)

AI-driven PdM research has grown substantially since 2021, fuelled by affordable sensors and maturing deep learning toolchains.

2.1 Dominant Methodologies

Recurrent deep learning dominates recent PdM literature; Bampoula et al. (2024) find Transformer encoders capture long-range dependencies better than LSTM autoencoders.

TCNs have also gained traction. Fahmi et al. (2024) introduce a TCN–Autoencoder with Multi-Head Attention for gas turbine fault detection, outperforming LSTM- and GRU-Autoencoder baselines (MSE 1.447, RMSE 1.193, MAE 0.712). TCNs use dilated causal convolutions enabling full parallelisation, a scalability advantage over LSTMs (Mao et al., 2025).

Elsherif et al. (2025) propose a CNN-LSTM hybrid achieving MAE 10.49–10.68 and PHM score 264.47–282.38 on C-MAPSS FD001.

2.2 Benchmark Datasets

The NASA C-MAPSS dataset remains the dominant RUL benchmark (Sharma et al., 2025), though it risks overfitting to turbofan patterns. The PRONOSTIA bearing dataset, from the IEEE PHM 2012 Challenge, provides run-to-failure bearing data under three load conditions. Kamat et al. (2021) achieve over 90% accuracy on PRONOSTIA using an Autoencoder-LSTM, making it the more relevant benchmark here. Elkateb et al. (2024) caution that both datasets omit the sensor drift, noise, and irregular sampling typical of live production environments.

2.3 Research Gaps

Three gaps are consistently acknowledged. First, labelled failure data are scarce, since machines are engineered not to fail, yielding severely imbalanced datasets (Elkateb et al., 2024). Second, cross-machine generalisation is poor, as most models are validated on a single equipment type and domain transfer remains largely unsolved (Bampoula et al., 2024). Third, real-time edge deployment is under-studied, as deep learning models are computationally heavy for embedded micro-controllers (Aly and Alotaibi, 2025; Aboshosha et al., 2023). This essay addresses gaps one and three through model choice and the deployment analysis in Section 6.

3. AI Model Selection and Justification

LSTM networks and TCNs are selected for the sensor data's high-frequency, multivariate nature and the dual tasks of fault classification and RUL regression.

3.1 Long Short-Term Memory (LSTM) Networks

LSTMs overcome the vanishing gradient problem via gating mechanisms, making them well-suited to sequential sensor data where temporal ordering carries diagnostic value (Bampoula et al., 2021). Its key limitation is sequential computation: inference latency and training cost scale unfavourably with window length.

3.2 Temporal Convolutional Networks (TCNs)

TCNs replace recurrence with dilated causal convolutions, capturing long-range dependencies while processing all time steps in parallel (Fahmi et al., 2024). Fahmi et al. (2024) and Mao et al. (2025) demonstrate competitive or superior accuracy versus LSTM variants, with faster training and CNN-native edge acceleration.

Table 1: Comparative Summary of Selected AI Models.

Dimension	LSTM	TCN
Architecture	Gated recurrent units; sequential	Dilated causal convolutions; parallel
Temporal range	Short–medium via gating	Long-range via dilation
Training speed	Slower (sequential)	Faster (parallelised)
Interpretability	Low (opaque gates)	Low (abstract filters)
Benchmark (RUL)	MAE 10.49–10.68, C-MAPSS FD001	MSE 1.447, gas turbine data
Edge deployability	Moderate (ONNX)	Good (CNN-native)

4. Data Requirements and Preprocessing

4.1 Data Requirements

Each machine class requires multivariate sensor streams: vibration, temperature, pressure, and acoustic emission. Vibration sampling must reach at least 10–25.6 kHz to resolve bearing fault frequencies (Elkateb et al., 2024). Where run-to-failure data are unavailable, PRONOSTIA and C-MAPSS serve as validated surrogates.

4.2 Handling Class Imbalance

Healthy observations vastly outnumber failure precursors, causing standard cross-entropy loss to default to the majority class. Elkateb et al. (2024) apply Borderline-SMOTE to synthesise minority-class samples in feature space. Cost-sensitive learning and Balanced Random Forests offer complementary strategies (Varalakshmi and Kumar, 2025).

4.3 Feature Extraction

Features span three domains: time-domain statistics, FFT-derived fault frequencies, and wavelet/STFT representations. For sequence models, a 30-step sliding window with 50% overlap generates training samples.

4.4 Data Splitting

Data are split 70/15/15% (train/validation/test), strictly respecting temporal order to prevent data leakage. A machine-stratified split, holding out entire machines for testing, better represents deployment on unseen equipment than random splitting.

Table 2: Recommended Preprocessing Pipeline.

Stage	Action	Rationale
1. Acquisition	Vibration, temperature, pressure, acoustic ≥ 10 kHz	Resolve bearing fault frequencies
2. Cleaning	Impute gaps; remove sensor dropouts	Maintain temporal continuity

Stage	Action	Rationale
3. Normalisation	Min–max or z-score per channel	Equalise feature magnitudes
4. Imbalance	Borderline-SMOTE or cost-sensitive loss	Prevent majority-class bias
5. Feature extraction	RMS, kurtosis, FFT, wavelet per window	Informative model inputs
6. Windowing	30-step window; 50% overlap	Maximise training samples
7. Splitting	70/15/15%, temporal + machine-stratified	Prevent leakage; test on unseen machines

5. Experimental Design

LSTM and TCN models are evaluated on fault classification and RUL regression using PRONOSTIA (primary) and C-MAPSS FD001/FD002 (secondary).

5.1 Evaluation Metrics

Classification is measured by precision, recall, F1 score, and AUC-ROC, with recall prioritised to minimise missed alarms. RUL regression uses RMSE, MAE, and the NASA PHM Prognostics Scoring Function which asymmetrically penalises late predictions as the domain-relevant metric (Elsherif et al., 2025). Confusion matrices and Wilcoxon signed-rank tests provide statistical validation.

5.2 Cross-Validation Strategy

Standard k-fold cross-validation is unsuitable for time series, so a walk-forward scheme of five chronological folds is used. Leave-one-machine-out (LOMO) evaluation additionally tests generalisation to unseen machines.

5.3 Comparison Criteria

Both models share identical preprocessing pipelines and data splits. Hyperparameters are tuned via Bayesian optimisation (50 trials) using the Adam optimiser with early stopping (max 200 epochs). Runs are replicated five times (mean $\pm$ SD reported).

Table 3: Experimental Design Summary.

Component	LSTM	TCN
Datasets	PRONOSTIA (primary); C-MAPSS FD001/FD002	PRONOSTIA (primary); C-MAPSS FD001/FD002
Tasks	Fault classification + RUL regression	Fault classification + RUL regression
Cross-validation	Walk-forward (5 folds) + LOMO	Walk-forward (5 folds) + LOMO
Metrics	Precision, Recall, F1, AUC-ROC, RMSE, MAE	Precision, Recall, F1, AUC-ROC, RMSE, MAE

Component	LSTM	TCN
Hyperparameter search	Bayesian optimisation (50 trials)	Bayesian optimisation (50 trials)
Replication	5 seeds; mean $\pm$ SD	5 seeds; mean $\pm$ SD
Ablation	LSTM vs. Bidirectional LSTM	TCN vs. TCN without attention

6. Critical Analysis: Interpretability, Scalability, and Deployment

6.1 Interpretability

Interpretability is critical in safety-sensitive manufacturing, where engineers must act on model outputs with confidence. Tree-based ensembles offer inherent transparency through feature importance scores mappable to engineering variables such as bearing temperature (Varalakshmi and Kumar, 2025). LSTM and TCN internal representations have no direct physical interpretation, embodying the classical deep learning black-box problem.

6.2 Scalability

LSTM's sequential computation scales poorly with fleet size, as training time grows super-linearly, demanding dedicated GPU infrastructure for periodic retraining. TCNs benefit from full parallelisation; their fixed receptive field also means memory does not grow with sequence length as LSTM hidden states do. Both support federated learning across edge devices, essential for multi-site deployments with bandwidth or privacy constraints (Aly and Alotaibi, 2025). Model versioning pipelines must include distribution-drift monitoring to detect silent degradation as operating conditions evolve (Elkateb et al., 2024).

6.3 Deployment

Live deployment requires sub-100ms inference latency for real-time pump and motor anomaly detection. TCN models export cleanly to ONNX and run efficiently on edge accelerators, whereas LSTM's stateful computation complicates on-device execution and typically requires INT8 quantisation (Aly and Alotaibi, 2025). Middleware translating AI predictions into maintenance work orders aligned with OPC-UA standards must be custom-built per vendor. Model drift, arising from sensor calibration changes or load shifts, demands a ground-truth feedback loop logging actual failures against predictions. Kullback–Leibler divergence drift detection with incremental fine-tuning mitigates catastrophic forgetting (Varalakshmi and Kumar, 2025).

7. Concluding Remarks

This essay has critically examined AI-based PdM for smart manufacturing, synthesising twelve sources from 2021–2025. Deep learning dominates the field, with C-MAPSS and PRONOSTIA as principal benchmarks. Key gaps remain: scarce labelled failure data, poor cross-machine generalisation, and under-addressed edge deployment challenges.

LSTM provides a well-validated baseline; TCN offers parallelism and edge-friendliness. Scalability favours TCNs, interpretability remains a shared limitation, and deployment demands drift monitoring and SCADA integration. A hybrid TCN–tree-based architecture represents the most pragmatic industrial solution. Future work should pursue domain adaptation for cross-machine transfer and federated learning for privacy-preserving multi-site deployment.

Acknowledgement

Perplexity AI was used in the preparation of this report to support literature searching, identify potentially relevant academic sources, and assist with grammar correction and word-limit reduction during editing. All sources included in the reference list were checked by the author and cited as the original academic works rather than citing Perplexity AI as a source. The analysis, interpretation, discussion, and final written content submitted in this report are the author's own work and responsibility.

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